Viva Preparation

Spatiotemporal Prediction of EV Charging Demand - questions and answers

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Course: Data Science Lab

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Dataset: UrbanEV - Li, H. et al. (2025), Scientific Data 12, 523. All figures and numbers in this document were generated from the project's executed code.

Project in one paragraph

We predict the energy (kWh) delivered by EV chargers in each of Shenzhen's 275 charging zones during the next hour, using the UrbanEV dataset (1,194,600 zone-hours, Sep 2022 - Feb 2023). We engineered 25 leakage-free features, compared Linear Regression, Random Forest and LightGBM on a chronological split, and selected **LightGBM** (test MAE 34.3 kWh, RMSE 142.7 kWh, R^2 0.972). It powers a Streamlit app for end users.

1. Dataset and its structure

Q. What dataset did you use?

A. UrbanEV (Li et al., Scientific Data, 2025): an open benchmark of EV charging in Shenzhen. We used the zone-level, cleaned, hourly version.

Q. How is the data structured?

A. Each hourly CSV is a wide table: a time column plus one column per zone (275 zones x 4,344 hours). We reshaped it into a long table with one row per zone-hour (1,194,600 rows).

Q. What is the target variable?

A. volume: total energy (kWh) delivered by chargers in a zone during an hour. It is continuous, so this is regression.

Q. Which other files did you use and how are they linked?

A. zone-information (by zone ID = TAZID), station information (station_id -> TAZID), pile rated power (pile -> station_id), and the 275 x 275 adjacency and distance matrices, whose order matches the zone columns. We verified every link.

Q. Were there missing values or data problems?

A. No missing, negative or duplicate values. We found zone 348 with (0, 0) coordinates (fixed from its stations), 20 one-way adjacency links (made symmetric) and 4 zones without neighbours (used the 3 nearest zones).

Q. Why did you not use the 5-minute data?

A. The hourly data is exactly the sum of the 5-minute data (max difference 0.000434 kWh), so it adds no new information for hourly forecasting and is 12 times larger. We used it only for this consistency check.

2. Regression

Q. Why is this a regression problem?

A. The output is a continuous quantity (kWh), not a category.

Q. What is supervised learning?

A. Learning a mapping from inputs to a known output using labelled examples; here each past zone-hour has its known demand as the label.

3. Linear Regression

Q. How does Linear Regression work?

A. It predicts $\hat{y} = b0 + b1 \cdot x1 + ... + bn \cdot xn$ and chooses the coefficients that minimise the sum of squared errors (ordinary least squares).

Q. Why did you scale features for Linear Regression only?

A. Scaling does not change tree splits, but for a linear model it puts coefficients on a comparable scale. The scaler was fitted on training data only, inside a pipeline.

Q. How do you interpret its coefficients?

A. With standardised features, a coefficient is the change in kWh for a one-standard-deviation increase in that feature. The largest is volume_lag24 (417.2 kWh). Because lag features are highly correlated, individual coefficients should be read with caution.

Q. Why did Linear Regression perform worst?

A. It can only fit straight-line relationships. Its error is high even on training data (RMSE 190.9 kWh), which indicates underfitting.

4. Random Forest

Q. How does a Random Forest work?

A. It trains many decision trees on random samples of rows, each split considering a random subset of features, and averages their predictions (bagging). Averaging reduces variance.

Q. Which hyperparameters did you use and why?

A. 100 trees, max depth 18, at least 20 samples per leaf, 50% of features per split and 30% of rows per tree. These limits keep training time and memory acceptable on an 8 GB laptop.

Q. How is feature importance calculated?

A. As the total reduction in squared error (impurity) contributed by each feature across all splits and trees, normalised to sum to 1.

5. LightGBM

Q. How does LightGBM work?

A. It is gradient boosting: trees are added one at a time, each fitted to the remaining errors of the current model and scaled by a learning rate. It grows trees leaf-wise and uses histograms of feature values, which makes it fast.

Q. What is the difference between bagging and boosting?

A. Bagging (Random Forest) trains trees independently and averages them, mainly reducing variance. Boosting (LightGBM) trains trees sequentially, each correcting the previous errors, mainly reducing bias.

Q. What is early stopping?

A. Training stops when the validation error has not improved for 100 rounds. LightGBM stopped at 369 trees, which prevents overfitting.

Q. Why did LightGBM perform best?

A. It captured non-linear patterns and interactions better than a linear model, and boosting with early stopping corrected errors without overfitting. Its test RMSE is 22.4% lower than Linear Regression and 11.9% lower than Random Forest.

6. Correlation and multicollinearity

Q. What does Pearson correlation measure?

A. The strength and direction of a LINEAR relationship between two variables, from -1 to +1. It does not imply causation.

Q. Which features correlate most with demand?

A. volume_lag24 (r = 0.96), volume_lag168 (0.95) and volume_lag1 (0.94). The only noticeable negative correlation is e_price (-0.23).

Q. Why does neighbour demand show almost zero correlation if it is useful?

A. Pooled over all zones r = -0.03, because size differences between zones dominate. Inside each zone over time the median r is 0.30.

Q. What is multicollinearity and did you find it?

A. Strong correlation between input features. We found 8 pairs with |r| >= 0.8, e.g. the demand lags with each other and occupancy_lag1 with duration_lag1. It makes linear coefficients unstable but does not harm tree models.

Q. Why is hour almost uncorrelated with demand?

A. Pearson only measures linear association; demand follows a daily cycle. The cyclical hour_cos feature has median within-zone r = 0.28.

7. Feature engineering and lag features

Q. What features did you create?

A. 25 inputs: demand lags (1, 24, 168 h), a 24-hour rolling mean, previous-hour occupancy and duration, neighbouring zones' previous-hour demand, tariff prices, calendar features (hour, day, weekday, weekend, holiday, sine/cosine cycles) and zone attributes (chargers, fast-charger share, power, area, location).

Q. What is a lag feature?

A. The value of a variable at an earlier time, e.g. volume_lag24 is the zone's demand 24 hours earlier. Lags were computed separately for each zone and shifted strictly into the past.

Q. Why lags of 1, 24 and 168 hours?

A. 1 h captures current momentum, 24 h captures the daily cycle and 168 h captures the weekly cycle.

Q. Why use sine/cosine for the hour?

A. Hour 23 and hour 0 are adjacent in time but far apart as numbers. Sine/cosine encoding places them close together.

Q. Why did you exclude month?

A. Training covers Sep-Dec and testing is February, so the model would see a month value it never learnt; six months cannot teach yearly seasonality.

8. Train/test splitting

Q. How did you split the data?

A. Chronologically: train 08 Sep 2022 - 31 Dec 2022 (759,000 rows), validation January 2023 (204,600), test February 2023 (184,800). All zones of a timestamp are in the same split.

Q. Why not a random split?

A. Random splitting would let the model train on hours after the test hours and on almost identical neighbouring hours, giving unrealistically good scores. Forecasting must always predict the future from the past.

Q. Why select the model on validation instead of test?

A. Choosing on the test set would make the reported test score optimistic. The test set is only used once, to confirm the choice.

9. Evaluation metrics

Q. Define MAE, MSE, RMSE and R^2.

A. MAE: average absolute error (kWh). MSE: average squared error (kWh^2), penalises large errors. RMSE: square root of MSE, back in kWh. R^2: share of variance explained, 1 is perfect.

Q. What do LightGBM's results mean in practice?

A. On average its forecast is 34.3 kWh away from the actual hourly demand of a zone (mean demand is about 261 kWh). RMSE 142.7 kWh is larger than MAE because a few large zones have larger errors. R^2 = 0.972 means it explains 97.2% of the variation.

Q. Is a high R^2 enough to say the model is good?

A. No. Part of R^2 comes from differences between big and small zones. That is why we compared against naive forecasts: LightGBM reduces RMSE by 50.3% compared with repeating the last hour.

10. Overfitting and underfitting

Q. What are overfitting and underfitting?

A. Overfitting: the model memorises training data and performs much worse on new data. Underfitting: the model is too simple and performs poorly even on training data.

Q. Did your models overfit?

A. Not noticeably. LightGBM: train RMSE 123.5, validation 141.4, test 142.7 kWh. Random Forest: 157.9 / 152.2 / 161.9. Linear Regression underfits (high error on training data).

Q. How did you prevent overfitting?

A. Chronological validation, early stopping for LightGBM, minimum samples per leaf and depth limits for Random Forest, and no tuning on the test set.

11. Model comparison

Model	MAE (kWh)	RMSE (kWh)	R^2
Linear Regression	48.70	183.85	0.9535
Random Forest	35.56	161.91	0.9639
LightGBM	34.28	142.67	0.9720
Naive (lag-1)	51.49	287.02	0.8867
Naive (lag-168)	77.19	323.78	0.8558

Q. Which features matter most?

A. For LightGBM: volume_lag24 (63.2%), volume_lag168 (20.5%), volume_lag1 (12.9%). Recent demand history dominates.

12. Data leakage

Q. What is data leakage?

A. Using information during training that would not be available at prediction time, which gives falsely good results.

Q. How did you avoid it?

A. Only past values (lags) of demand, occupancy and duration are used; volume_11kw (part of the target) is excluded; the scaler is fitted on training data only; the split is chronological; and lag values were checked automatically against the raw data.

Q. Why can you use same-hour prices but not same-hour occupancy?

A. Prices follow a published time-of-use tariff (97% of days repeat the previous day's profile), so they are known in advance. Occupancy is only known after the hour has happened.

Q. Why is volume_11kw leakage?

A. It is the energy from slow chargers, which is part of the total volume (≤ volume in 99.4% of rows, 39% of all energy). Using it means partly giving the model the answer.

13. Walk-forward forecasting and future dates

Q. What is walk-forward forecasting?

A. Each month is predicted by models trained only on the months before it (train Sep -> predict Oct, train Sep-Oct -> predict Nov, ... train Sep-Jan -> predict Feb). It simulates retraining the model every month as new data arrives.

Q. Why did you use it?

A. So users can pick any date from 1 Oct 2022 to 28 Feb 2023 and compare an honest prediction with the actual demand, and to check that the model ranking is stable over time. Over Oct-Feb LightGBM reached RMSE 154.2 kWh (R^2 0.966), versus 168.6 for Random Forest and 186.4 for Linear Regression.

Q. Why can't September be predicted?

A. It is the first month of the data, so there is no earlier data to train on. Predicting it with a model that learnt from it would not be an honest forecast.

Q. How can you predict dates after February 2023?

A. Next-hour models need the last hours' real demand, which does not exist after the dataset. For later dates we use profile models: inputs known for any date (hour, weekday, holiday, typical prices, zone attributes and the zone's typical demand), trained on all six months.

Q. How accurate are the future-date estimates?

A. Tested on February: best model Linear Regression, RMSE 301.9 kWh, only 2.1% better than just using the zone's typical demand (308.2). Next-hour forecasts on the same month are far better (140.0). Recent demand is the key information.

Q. Why is Linear Regression best for future dates when LightGBM is best overall?

A. Without recent demand the task becomes 'scale the zone's usual pattern'. The tree models learnt Sep-Jan specifics that did not carry over to February, while the simpler linear model generalised better. The best model depends on the task and the inputs.

Q. What are the limits of future estimates?

A. They reflect Sep 2022 - Feb 2023 patterns only: no summer months, no growth in EV numbers or new stations after February 2023, and they cannot be checked against actual data. The app labels them as estimates.

14. Application and practical use

Q. How does the web application work?

A. It is a Streamlit app. The user picks a zone and any date from 1 Oct 2022 to today. For dataset dates it shows all three models' predictions next to the actual demand with an accuracy label; for later dates it shows estimates with a typical error range. Pages: Home, Demand Forecast, Prediction vs Actual, Demand Map, Zone Insights, Compare Zones, About the Model.

Q. What are the practical applications?

A. Operators: capacity planning, dynamic pricing, maintenance in quiet hours. Drivers: avoiding queues. Planners: identifying overloaded zones. Grid operators: anticipating EV charging load.

Q. What did you find surprising?

A. Demand peaks overnight: 46% of energy is delivered between 00:00 and 07:59, when electricity is cheapest, consistent with electric taxis and ride-hailing fleets charging at night.

Q. What are the main limitations and improvements?

A. Six months of data, one city, no weather data, one-hour horizon. Improvements: weather/event data, multi-step forecasts, graph neural networks using the adjacency matrix, and live data.